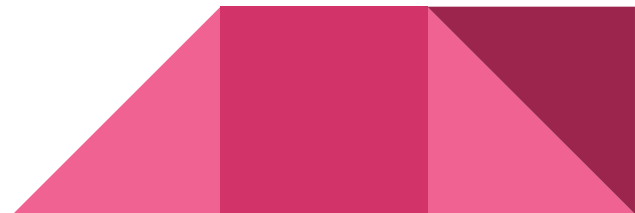


Differentially Private DRO

-Shruthi & Sushanth

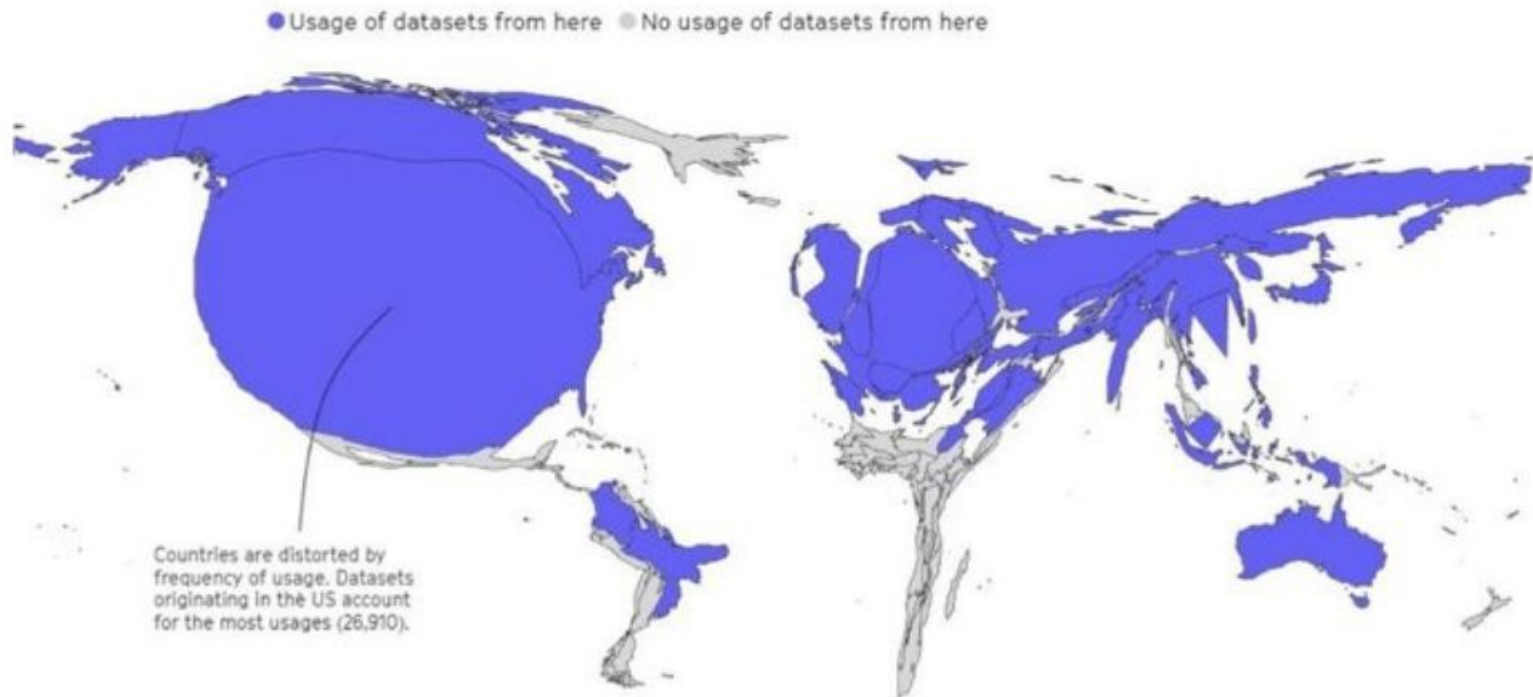
Contents

1. Introduction
2. Some Theoretical Results
3. Experiments



Introduction to DR0

AI builds on data as infrastructure



Distributional Shift from train to test

Pattern recognition will reflect existing biases



Screenshot from 2020-03-31 11-27-22.png

Technology	68%
Electronic Device	66%
Photography	62%
Mobile Phone	54%



Screenshot from 2020-03-31 11-23-45.png

Gun	88%
Photography	68%
Firearm	65%
Plant	59%

Distributional Shift, Generalizability and Group Bias

- A dataset *distributional shift* occurs when the statistical properties of the training data differ from those of the test or deployment data. This can cause models trained on historical data to perform poorly when applied to new, unseen data.
 - Model *generalizability* refers to a model's ability to perform well on new, unseen data that was not part of the training set. Generalizability is crucial because it indicates that a model has captured the underlying structure and patterns of the data rather than just memorizing the training examples.
 - Minority *group bias* occurs when a model performs disproportionately poorly on underrepresented groups in the data (e.g., based on race, gender, age, etc.). This can lead to biased predictions that negatively impact these minority groups.
- 

Distributionally Robust Learning/Optimization (DRO)

- Focuses on Reducing the Worst Case Losses rather than the average Loss, by giving more importance(weightage) to higher loss.
- In this case the minority groups whose loss might be high will be worked upon more.

We can thus consider the objective of the form

$$R_D(\theta) := \max_{p \in P} \sum_{i=1}^n p_i l_i(\theta) - \nu h(p)$$



Importance of DP with DRO in Healthcare

Why DRO: Healthcare models need to be robust across diverse patient populations, where training and test data distributions can vary significantly due to factors like age, ethnicity, or geographic location. DRO helps models generalize across these groups, making them more reliable for various patient demographics.

Why Privacy: Health data is highly sensitive and regulated by laws like HIPAA in the U.S. and GDPR in Europe. Protecting patient privacy is essential, and data handling must adhere to stringent security standards.

DRO and Privacy Together: Privacy-preserving techniques, such as differential privacy, can be combined with DRO to ensure that models are robust without exposing individual patient data.



The Conflict

- DP aims to minimize the impact of individual data points on the model's output, thereby enhancing privacy
- DRO seeks to optimize performance across worst-case distributions often amplifying the influence of minority groups or rare cases, possibly leading to leakage of information of minority groups
- There is therefore a tradeoff between privacy and robustness / fairness.





Theory

How sensitivity changes for DRO algorithms?

$$R_D(\theta) := \max_{p \in P} \sum_{i=1}^n p_i l_i(\theta) - \nu h(p)$$

For a specific dataset D , this becomes:

$$\nabla R_D(\theta) = p_D^{*T} \nabla_{\theta} l_D(\theta) \quad p_D^*(\theta) := \arg \max_{p \in P} \sum_{i=1}^n p_i l_i(\theta) - \nu h(p)$$

The sensitivity Δ is defined as follows:

$$\Delta = \max_{D \simeq D'} \|\nabla_{\theta} R_D(\theta) - \nabla_{\theta} R_{D'}(\theta)\|_2$$

Finally, we conclude:

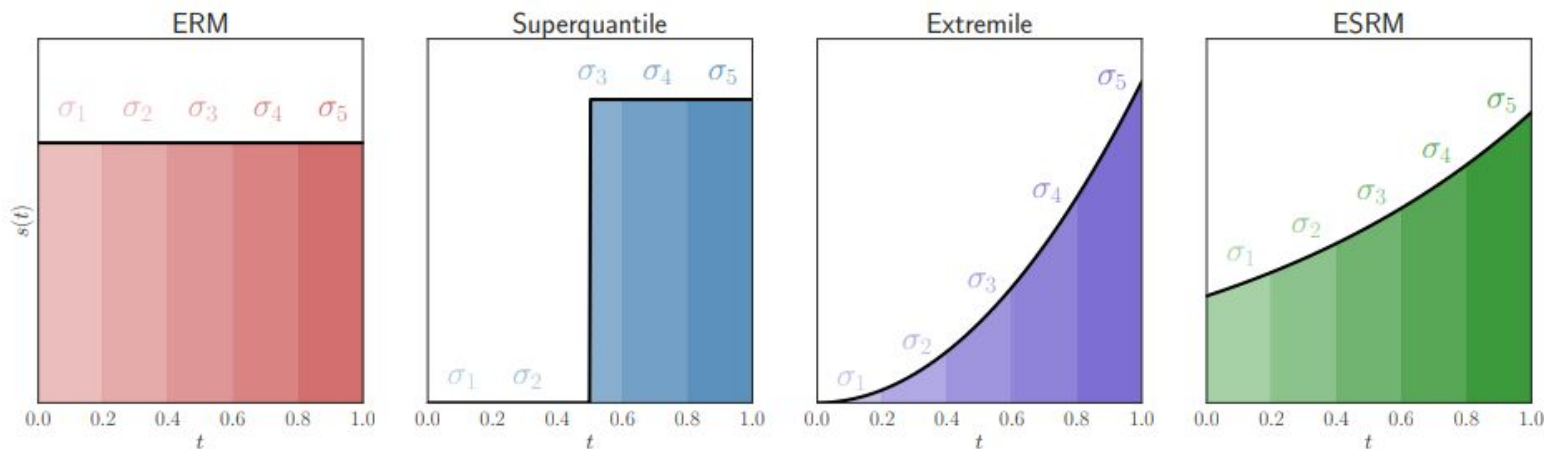
$$\Delta \leq \left(\min\left(2, \frac{B}{\nu}\right) + 1 \right) C \quad B = \max l_D(\theta, u_n)$$



Spectral Risks Method

In this method, p is a permutation of chosen fixed constants $\sigma_1 \leq \sigma_2 \leq \dots \leq \sigma_n$ where $\sum_{i=1}^n \sigma_i = 1$, where data points are sorted in ascending order of loss values. Hence $l_1(\theta) \leq l_2(\theta) \dots l_n(\theta)$. Thus,

$$R_D(\theta) = \sum_{i=1}^n \sigma_i l_{Di}(\theta)$$



Sensitivity for Spectral Risks

$$\nabla R_D(\theta) = \sum_{i=1}^n \sigma_i \nabla l_{Di}(\theta)$$

and

$$\nabla R_{D'}(\theta) = \sum_{i=1}^{n-1} \sigma_{i+1} \nabla l_{Di}(\theta)$$

Therefore,

$$\begin{aligned} \nabla R_D(\theta) - \nabla R_{D'}(\theta) &= \sum_{i=1}^{n-1} (\sigma_{i+1} - \sigma_i) \nabla l_{Di}(\theta) + \sigma_n \nabla l_{Dn}(\theta) \\ \Delta &\leq \sum_{i=1}^{n-1} |\sigma_{i+1} - \sigma_i| \|\nabla l_{Di}(\theta)\|_2 + |\sigma_n| \|\nabla l_{Dn}(\theta)\|_2 \leq C \left(\sum_{i=1}^{n-1} (\sigma_{i+1} - \sigma_i) + \sigma_n \right) \\ &\Rightarrow \Delta \leq C(2\sigma_n - \sigma_1) \end{aligned}$$

Experiments

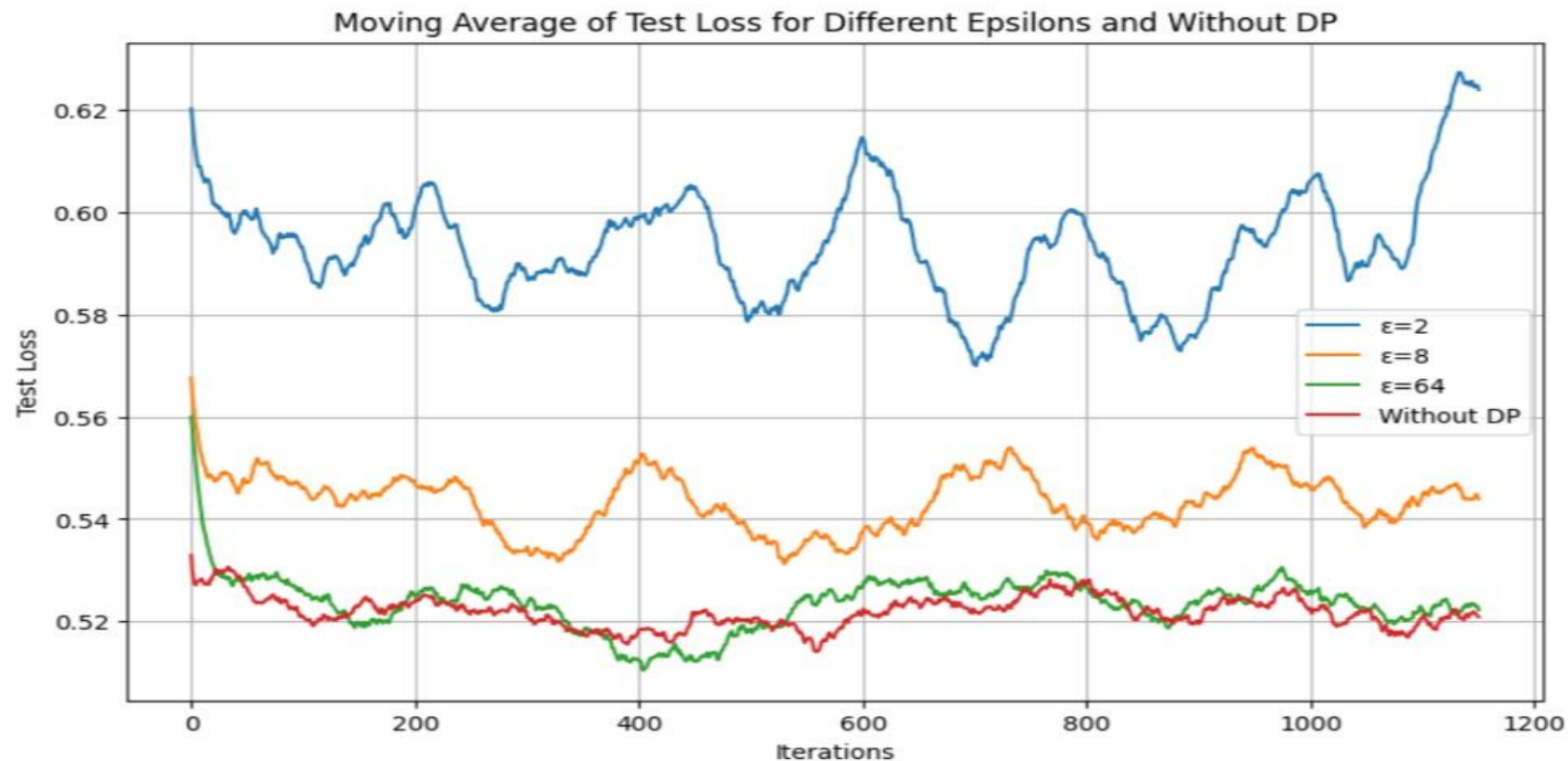
Suboptimality

$$\text{suboptimality gap}_t := \frac{\mathcal{R}_\sigma(w^{(t)}) - \mathcal{R}_\sigma(w^*)}{\mathcal{R}_\sigma(w^{(0)}) - \mathcal{R}_\sigma(w^*)}$$

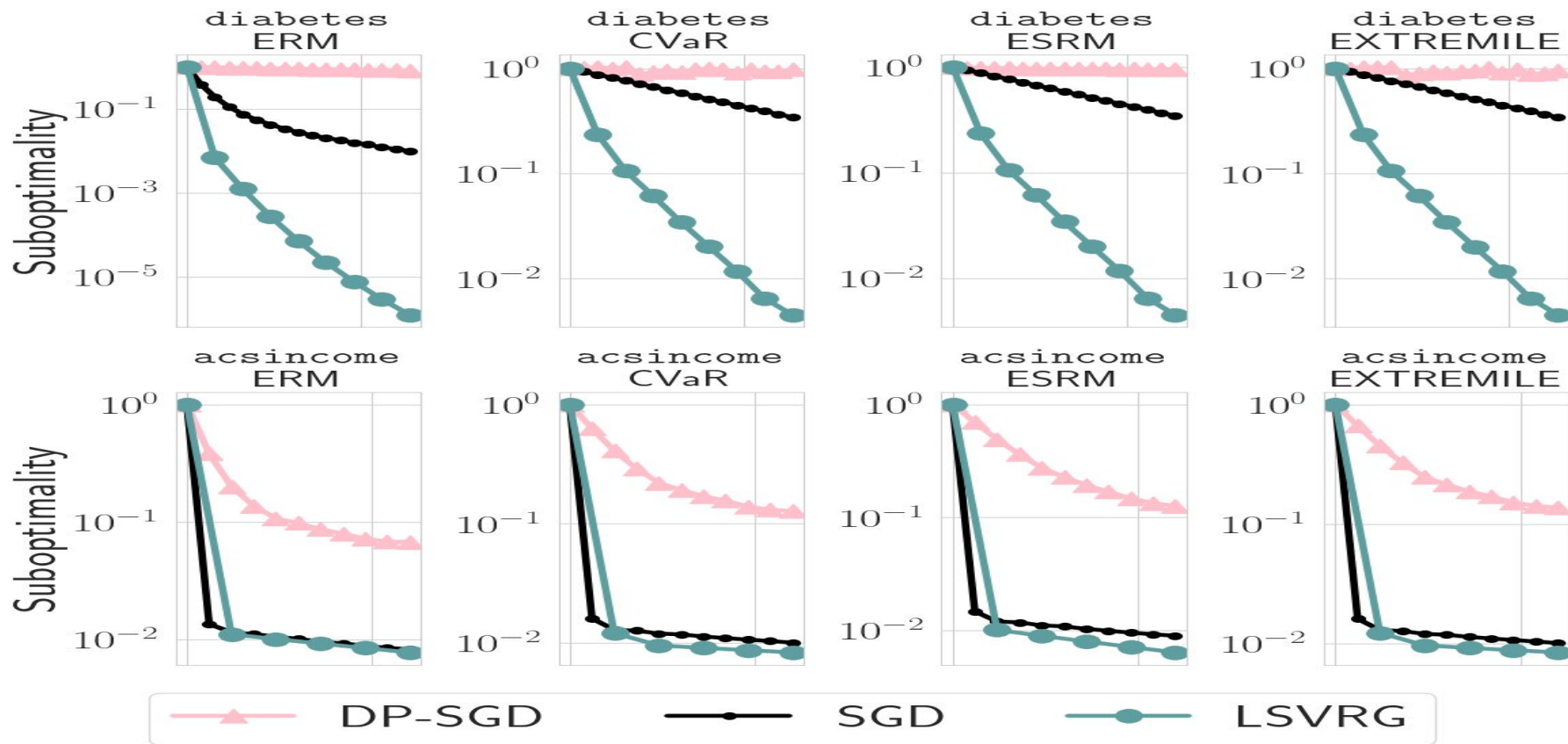
- The plot of suboptimality gives an insight into how the target function changes across iterations.
- Since our target function focuses on fairness/minority groups- suboptimal plots could be used to compare model performance.



Performance of DP-SGD and SGD



Comparison of Suboptimality



Future Work

- Finding the noise-multiplier for Differentially Private LSVRG algorithm (L-risk Stochastic Variance-Reduced Gradient)
- Implementation of DP-LSVRG and comparing its performance.
- Exploring other fairness metrics and benchmarking.

